

5.7 Field data, statistical tests

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When the use of field data is needed, and it concerns newly occurred diseases, only families with affected offspring can be considered. When a segregation ratio for Mendelian inheritance is calculated on the base of such observations, it is necessary to correct for families that due to random chances are not segregating.

If recessive inheritance occurs and both parents in a family are heterozygotes the chance of the family not being detected, by means of an affected first born, is $3/4$. After the birth of the second offspring the chances will be $(3/4)^2 = 9/16$. Figure 5.7 shows how heterozygotic parents will segregate in a family of two. 9 out of 16 families will have no affected offspring, so by excluding these families in the final analysis, the classical Mendelian analysis becomes meaningless.

In the following are formulas for testing and calculating the corrected segregation in families which have chances of segregation. Therefore the relevant data (shaded area) from Figure 5.7 is:

T = 14 -total number
 A = 8 -number of affected
 $A_1 = 6$ -number of families with 1 affected
 $A_2 = 1$ -number of families with 2 affected

Figure 5.7
Theoretical segregation of heterozygotic parents in 7 out of 16 families of two.

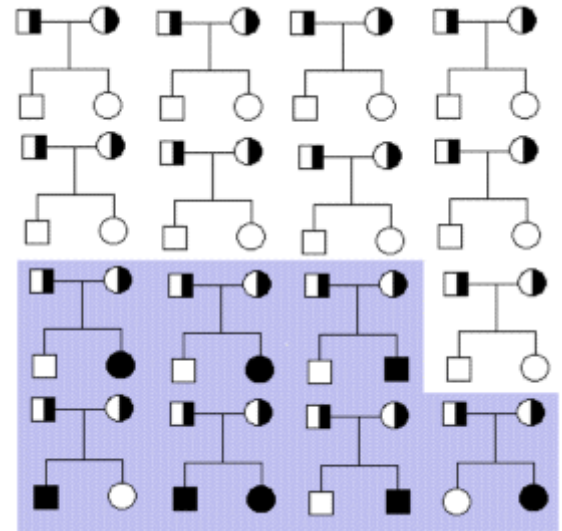
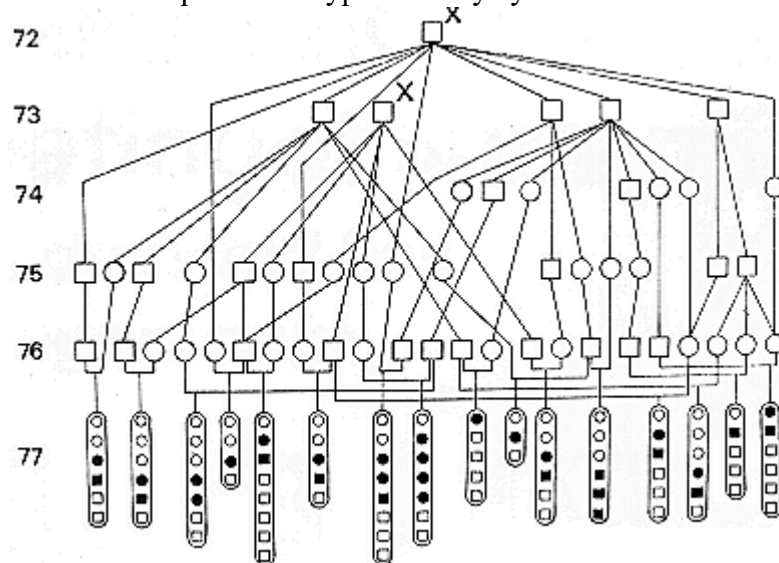


Figure 5.8 shows a path diagram of a mink family from a farm with an outbreak of tyrosinaemia. One of the clinical symptoms is, as shown in the picture, the fact that the eyelids stick together. The whelps die after one or two days with clinical symptoms. The cases occur when the whelp is about six weeks. A description of the disease can be found in *et al. Canadian J. Comparative Medicine*, 43:333-340, 1979.

Figure 5.8. Segregation of tyrosinaemia in standard mink (filled symbols), The photo on the right shows an affected whelp with the typical sticky eyelids.



The formulas to apply according to "Singles method" for a test of Mendelian inheritance (for the proportion $p\text{-hat}$) are shown below. The actual numbers from the tyrosinaemia case, Figure 5.8, are inserted.

$$\begin{aligned}\hat{p} &= \frac{A - A_1}{T - A_1} = \frac{32-4}{94-4} = 0,311 \\ \text{Var}(\hat{p}) &= \frac{T - A}{(T - A_1)^3} \left[A - A_1 + 2A_2 \frac{T - A}{T - A_1} \right] \\ &= \frac{94-32}{(94-4)^3} \left[32-4 + 2 \cdot 9 \frac{94-32}{94-4} \right] \\ &= 0,0034 \\ Z^2 &= \frac{(\hat{p} - p)^2}{\text{Var}(\hat{p})} = \frac{(0,311-0,25)^2}{0,0034} = 1,10\end{aligned}$$

T and A are respectively the total number of offspring and the number of affected offspring. A_1 and A_2 are the number of families with respectively 1 and 2 affected offspring. Z^2 is Chi-squared distributed with 1 degree of freedom. $\hat{p}$ is tested against the expected segregation frequency (p) which in the tyrosinaemia case is 0.25, as it is expected that both parents are carriers.

In the mink family is found that:

T = 94 -total number
A = 32 -number of affected
 A_1 = 4 -number of families with 1 affected
 A_2 = 9 -number of families with 2 affected

The calculation is carried out by inserting the actual numbers in the formulas above. There is no significant statistical deviation from a 3:1 segregation ratio as the test value is less than 3,84 ($H_0: \hat{p} = p = 0,25$).

Families with more than 2 affected individuals are not directly part of the calculations, but contribute only to the total number of affected.

An applet for the calculation can be found [here](#)

If the number from the first theoretical example (Figure 5.7) is inserted in the formula for $\hat{p}$ we get $(8-6)/(14-6) = 0.25$, which is of course the correct result, as the data originated from mating with known heterozygotes.

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